Business Text Mining

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Text Pre-processing
Part-of-Speech Tagging

Natural Language Processing

- **Lexical Analysis**
  - Tokenization, Morphological Analysis
  - Part-of-Speech (POS) Tagging
  - Additional analysis: Named Entity Recognition (NER), noun phrase recognition, sentence split, etc
- Syntax Analysis
- Semantic Analysis

Part-of-Speech Tagging

- Assigning a Part-of-Speech to each word in a text
- Words often have more than one POS tag
- e.g., **back**
  - Earnings growth took a back/ADJ seat
  - A small building in the back/NOUN
  - A clear majority of senators back/VERB the bill
  - Enable the country to buy back/PART debt
  - I was twenty-one back/ADV then

I	like	his	watch
PRO	VERB	PRO	NOUN

The	man	fans	the	flame
DT	NOUN	VERB	DT	NOUN

The	fans	watch	the	race
DT	NOUN	VERB	DT	NOUN

Two Classes of Words: Open vs. Closed

- Closed class words
  - Relatively fixed membership
  - Usually function words: short, frequent words with grammatical function
    - Determiners: *a, an, the*
    - Pronouns: *she, he, I*
    - Prepositions: *on, under, over, near, by,…*

Two Classes of Words: Open vs. Closed

- Open class words
  - Usually content words:
    - Nouns, Verbs, Adjectives, Adverbs
    - Interjections: *oh, ouch, uh-huh, yes, hello*
  - New Nouns and Verbs like *iPhone* or *google it*

Open class ("content") words

Nouns

- Proper: *Janet* *Italy*
- Common: *cat, cats* *mango*

Verbs

- Main: *eat* *went*
- Auxiliary: *can* *had*

Adjectives *old green tasty*

Adverbs *slowly yesterday*

Numbers *122,312* *one*

Interjections *Ow hello*

… more

Closed class ("function")

- Determiners *the some*
- Conjunctions *and or*
- Pronouns *they its*
- Prepositions *to with*
- Particles *off up*

… more

한글 형태소 품사 태그표 (Korean POS Tagger)

대분류	세종 품사 태그		심광섭 품사 태그		KKMA 단일 태그 V 1.0					
	태그	설명	Class	설명	묶음 1	묶음 2	태그	설명	확률태그	저장사전
체언	NNG	일반 명사	NN	명사	N	NN	NNG	보통 명사	NNA	noun.dic
	NNP	고유 명사					NNP	고유 명사		
	NNB	의존 명사	NX	의존 명사			NNB	일반 의존 명사	NNB	simple.dic
			UM	단위 명사			NNM	단위 의존 명사		
	NR	수사	NU	수사		NR	NR	수사	NR	
	NP	대명사	NP	대명사		NP	NP	대명사	NP	
용언	VV	동사	VV	동사	V	VV	VV	동사	VV	verb.dic
	VA	형용사	AJ	형용사		VA	VA	형용사	VA	
	VX	보조 용언	VX	보조 동사		VX	VXV	보조 동사	VX	
			AX	보조 형용사			VXA	보조 형용사		
	VCP	긍정 지정사	CP	서술격 조사 '이다'		VC	VCP	긍정 지정사, 서술격 조사 '이다'	VCP	raw.dic
	VCN	부정 지정사					VCN	부정 지정사, 형용사 '아니다'	VCN	
관형사	MM	관형사	DT	일반 관형사	M	MD	MDT	일반 관형사	MD	simple.dic
			DN	수 관형사			MDN	수 관형사		
부사	MAG	일반 부사	AD	부사		MA	MAG	일반 부사	MAG	
	MAJ	접속 부사					MAC	접속 부사	MAC	
감탄사	IC	감탄사	EX	감탄사	I	IC	IC	감탄사	IC	

Source: http://kkma.snu.ac.kr/documents/?doc=postag

Different POS Tags for the Same Token

```
In [2]: phrase = 'I am meeting him tomorrow at the meeting'
        doc = nlp(phrase)

        for token in doc:
            print(f'{token.text:{15}} {token.lemma_:{15}} {token.lemma:<{25}} {token.pos_:{10}}')
```

```
I               I               4690420944186131903       PRON
am              be              10382539506755952630      AUX
meeting         meet            6880656908171229526       VERB
him             he              16553127710671082 81      PRON
tomorrow        tomorrow        3573583789758258062       NOUN
at              at              11667289587015813222      ADP
the             the             7425985699627899538       DET
meeting         meeting         14798207169164081740      NOUN
```

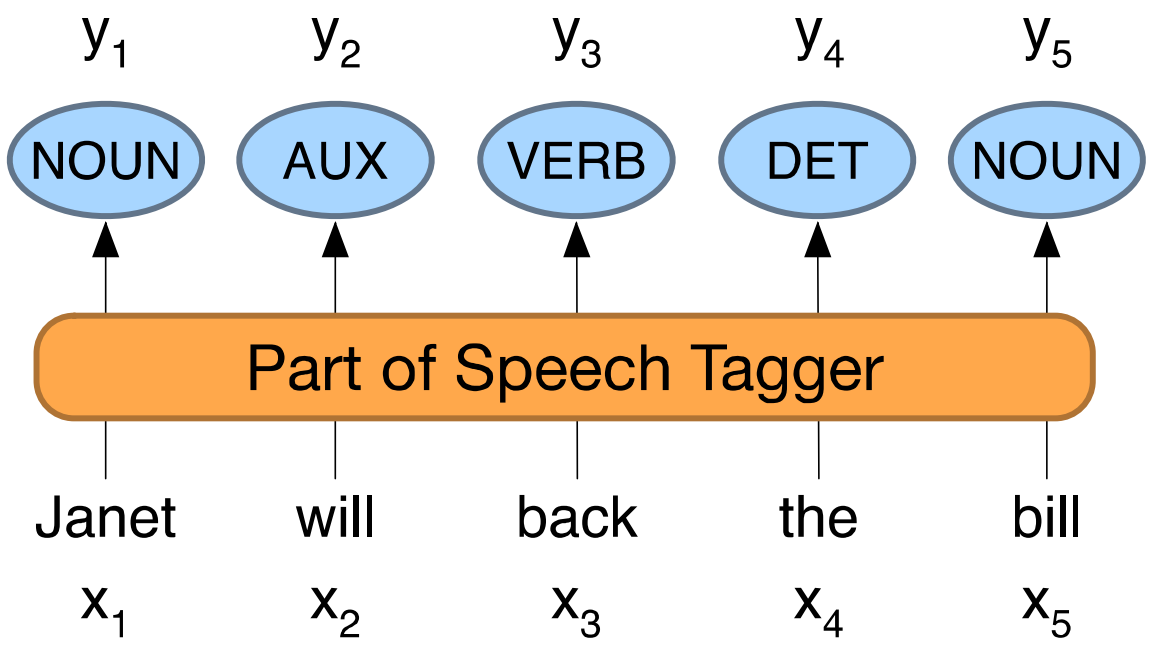
Lexical Analysis
Part-of-Speech (POS) Tagging

- Given a sentence X, predicts its POS sequence Y
  - **Input**: a sequence of tokenized words
  - **Output**: a sequence of corresponding tags
- E.g.,

A	dog	is	chasing	a	boy	on	the	playground
↓	↓	↓	↓	↓	↓	↓	↓	↓
DT	NOUN	AUX	VERB	DET	NOUN	PERP	DT	NOUN

Part-of-Speech (POS) Tagging

- A statistical POS tagger scans tokens and assigns tags



  - Relies on different word order probabilities
  - Needs a manually tagged corpus for machine learning

POS Tagging

- POS Tagging generally requires:
  - **Training phase** where a manually annotated corpus is processed by a machine learning algorithm
  - **Tagging algorithm** that processes texts using learned parameters
  - Performance is generally good (around 96%) when staying in the same domain

POS Tagging (cont')

- Algorithms used in POS Tagging :
  - Hidden Markov Models (HMMs)
    - find the most probable tag sequence given the sentence
  - Decision trees
  - Support Vector Machines (SVM)

POS Tagging (cont')

- Probabilistic Model for POS Tagging
  - Find the *most probable tag sequence given the sentence*

A	dog	is	chasing	a	boy	on	the	playground
↓	↓	↓	↓	↓	↓	↓	↓	↓
DT	NOUN	AUX	VERB	DET	NOUN	PERP	DT	NOUN

$$\underset{Y}{\operatorname{argmax}}\, P(Y|X)$$

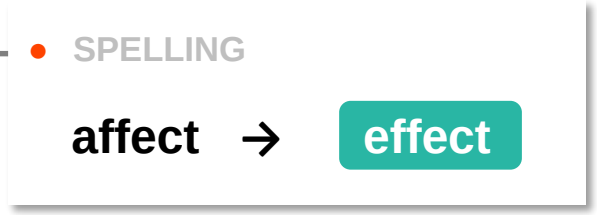
POS Tagging (cont')

- Generative Sequence Model
  - Decompose probability using Bayes' Rule

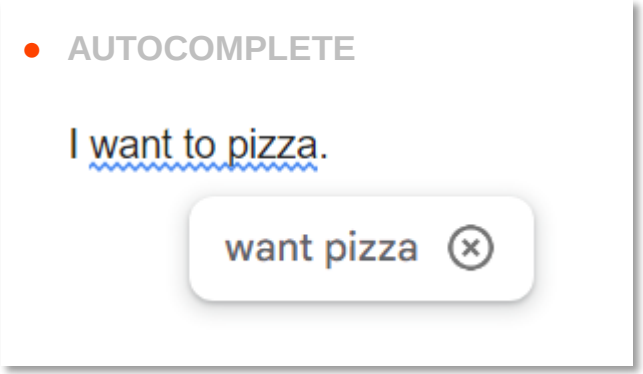
$$\begin{aligned}\underset{Y}{\operatorname{argmax}}\, P(Y|X) &= \underset{Y}{\operatorname{argmax}} \frac{P(X|Y)\,P(Y)}{P(X)} \\ &= \underset{Y}{\operatorname{argmax}}\, P(X|Y)\;P(Y)\end{aligned}$$

Applications of POS Tagging

- Text-to-Speech recognition system
  - How do we pronounce “lead” or “object”?
- Spelling correction
  - “This announcement is going to have a significant affect on everyone’s outlook.”



- Autocomplete
  - “I want to pizza”



Applications of POS Tagging

- Machine translation
  - Earnings growth took a back/ADJ seat → 이익 성장이 뒷전으로 밀렸다
  - A small building in the back/NOUN → 뒤쪽의 작은 건물
  - A clear majority of senators back/VERB the bill → 상원의원의 대다수가 법안을 지지한다
  - Enable the country to buy back/PART debt → 국가가 부채를 다시 사들일 수 있게 해주세요
  - I was twenty-one back/ADV then → 그때 내 나이 스물한 살이었어

Applications of POS Tagging

- Features in Text Modeling

By 1925 present-day Vietnam was divided into three parts under French colonial rule. The southern region embracing Saigon and the Mekong delta was the colony of Cochin-China; the central area with its imperial capital at Hue was the protectorate of Annam…

Clara never failed to be astonished by the extraordinary felicity of her own name. she found it hard to trust herself to the mercy of fate, which had managed over the years to convert her greatest shame into one of her greatest assets.

Source: Argamon, S., Koppel, M., Fine, J., & Shimoni, A. R. (2003). Gender, genre, and writing style in formal written texts. Text & talk, 23(3), 321-346.

Applications of POS Tagging

In [44]:

```
# 부정적 리뷰: 평점 1, 2점
review_neg = dataset[(dataset['score'] == 1) | (dataset['score'] == 2)]['review_prep']

# 명사만 추출하는 함수
def extract_nouns(text):
    doc = nlp(text)
    nouns = [token.text for token in doc if token.pos_ in ['NOUN', 'PROPN']]
    return nouns

# 각 review_pos에 대해 명사 추출하여 review_tokenized_pos에 저장
review_tokenized_neg = [extract_nouns(review) for review in review_neg]
```

https://github.com/hoooooooni/google_review_LDA_english/blob/main/LDA_Lemmatizer.ipynb